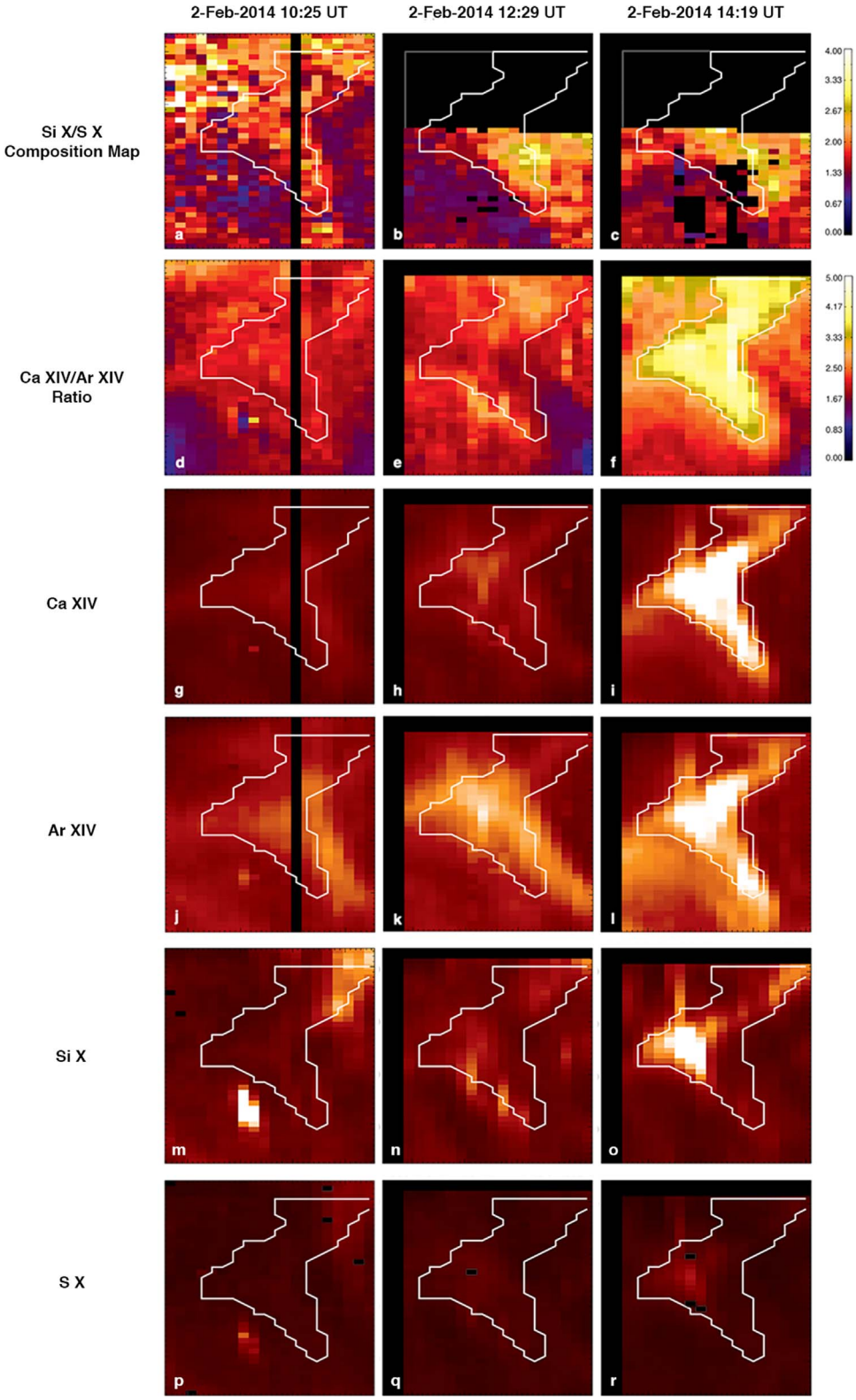

Figure 3. Top to bottom: Hinode/EIS Si X 258.38 Å/S X 264.23 Å composition map, Ca XIV 193.87 Å/Ar XIV 194.40 Å intensity ratio maps, Ca XIV 193.87 Å intensity maps, Ar XIV 194.40 Å intensity maps, Si X 258.38 Å intensity maps and S X 264.23 Å intensity maps. The contour indicates the flaring region used for analysis. Left to right: observations from three different EIS spectral scans on 2014 February 2, starting at 10:25 UT, 12:29 UT, and 14:19 UT, respectively. In the composition and intensity ratio maps, blue represents a photospheric-like (unenhanced composition) FIP bias of 1; red represents a quiet-Sun composition and yellow shows a high-FIP bias composition. A clear X shape of the structure could be observed in the 14:19 UT raster. The algorithm to calculate the Si X 258.38 Å/S X 264.23 Å composition map minimizes temperature and density effects by utilizing both the short-wavelength and long-wavelength detectors of Hinode/EIS. Offset between the two detectors contributed to the patch of dark pixels in the composition maps; the Ca XIV 193.87 Å/Ar XIV 194.40 Å ratio maps are simply the intensity ratio of the two spectral lines. A more detailed description can be found in Sections 2.3.1 and 2.3.2.